User Stories & Acceptance Criteria

EV-Optimized Route Planning Platform

EPIC	USER STORY	ACCEPTANCE CRITERIA
As an EV User, I want to register or log in securely so that I can access personalized trip planning, vehicle information, and trip history. *Source: User Registration / Login*	*As a new EV User, I need to create an account with my name, email, phone number, and password **so that** I can use the platform.*	Ensure the EV User is able to: • Enter name, email, phone number, and password. • Have the system validate email uniqueness and password strength. • Have an account created successfully. • Be redirected to the dashboard after registration.
	*As a registered EV User, I need to log in using my email and password **so that** I can access my account and previous trips.*	Ensure the EV User is able to: • Enter valid login credentials. • Receive an error for incorrect credentials. • Retry login if authentication fails. • Access the home page after successful login.
As an EV User, I want to enter my trip details so that the system can generate the most battery-efficient route. *Source: Trip Planning*	*As an EV User, I need to enter my current location, destination, vehicle model, and battery percentage **so that** the system can calculate the best route.*	Ensure the EV User is able to: • Enter source and destination. • Select vehicle type and model. • Enter current battery percentage. • Submit trip details successfully.
As an EV User, I want to receive battery-efficient route recommendations so that I can minimize battery consumption during my journey. *Source: Battery-Optimized Route Generation*	*As an EV User, I need the system to generate multiple routes and recommend the most battery-efficient route **so that** I can travel efficiently.*	Ensure the EV User is able to: • View multiple route options. • View estimated distance and travel time. • View battery consumption for each route. • Receive the recommended battery-efficient route.
As an EV User, I want to know my expected battery usage before starting the trip so that I can travel confidently. *Source: Battery Usage Prediction*	*As an EV User, I need the system to predict battery consumption using route distance, traffic, weather, and terrain information.*	Ensure the EV User is able to: • View estimated battery usage. • View remaining battery percentage. • Receive battery predictions before starting the trip. • Receive updated predictions when conditions change.

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As an EV User, I want charging station recommendations so that I can recharge my vehicle whenever necessary. *Source: Charging Station Recommendation*	*As an EV User, I need to locate nearby charging stations along my route **so that** I can safely complete my journey.*	Ensure the EV User is able to: • View nearby charging stations. • View charging station location. • View estimated distance to each charging station. • Select a charging station for navigation.
As an EV User, I want the platform to consider weather and terrain conditions so that battery consumption is estimated accurately. *Source: Weather & Terrain Impact*	*As an EV User, I need battery predictions to change automatically based on weather and road elevation.*	Ensure the EV User is able to: • View weather conditions. • View terrain impact. • Receive updated battery estimates. • Receive an updated optimized route when required.
As an EV User, I want real-time traffic monitoring so that I can avoid delays and unnecessary battery usage. *Source: Real-Time Traffic Monitoring*	*As an EV User, I need live traffic updates and alternate route suggestions during my journey.*	Ensure the EV User is able to: • View live traffic conditions. • Receive alternate route suggestions. • View updated travel time. • Continue navigation using the optimized route.
As an EV User, I want a complete trip summary so that I can review my journey before travelling. *Source: Trip Summary*	*As an EV User, I need to view route details including battery usage, travel time, distance, and charging stops.*	Ensure the EV User is able to: • View total distance. • View estimated travel time. • View battery consumption. • View remaining battery percentage. • View recommended charging stops.
As an EV User, I want to review my previous journeys so that I can analyze my travel and battery usage. *Source: Trip History*	*As an EV User, I need to access my trip history **so that** I can review previous routes and battery consumption.*	Ensure the EV User is able to: • View previous trips. • View travel date and destination. • View battery usage for each trip. • Delete or archive previous trips if required.

EPIC	USER STORY	ACCEPTANCE CRITERIA
As an Administrator, I want to manage vehicle and charging station information so that users always receive accurate route recommendations. *Source: Vehicle & Charging Station Management*	*As an Administrator, I need to add, edit, and delete EV models and charging station information.*	Ensure the Administrator is able to: • Add EV models. • Update battery specifications. • Add, edit, or remove charging stations. • Save updated system information successfully.
As an Administrator, I want to monitor platform usage so that the system operates efficiently. *Source: Platform Monitoring & Reporting*	*As an Administrator, I need to manage user accounts and monitor system activities.*	Ensure the Administrator is able to: • View registered users. • Manage user accounts. • Monitor trip requests. • View system reports and activity logs.